

REPORT

Grammar Proficiency Prediction from Spoken English Audio

1. Introduction

This project addresses the problem of predicting the grammar proficiency of spoken English audio samples. Given a short recording of a speaker, the goal is to estimate a numeric grammar score that reflects how accurately and fluently the speaker uses English grammar.

2. Approach

The solution is a pipeline that converts raw speech into a predicted score:

Audio	Whisper Transcription	Feature Extraction	Random Forest	Grammar Score
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Three groups of features are used:

- **Audio features** – acoustic properties of the recording.
- **Text features** – statistics of the Whisper transcript.
- **Grammar/POS features** – linguistic structure derived with spaCy.

3. Preprocessing

- Input data consists of **.wav** audio files.
- Speech is transcribed to text using **Whisper**.
- Transcripts undergo **text cleaning and tokenization**.
- **spaCy** is used for linguistic analysis (POS tags, tense, dependency parsing).
- **Missing-value and file validation** checks ensure every sample has a valid audio file and complete features.

4. Feature Engineering

- **Audio:** duration, RMS energy, zero-crossing rate (ZCR) and MFCCs.
- **Text:** word count and sentence count.
- **Sentence structure:** average sentence length.
- **Lexical:** vocabulary richness and repetition ratios.
- **Grammar:** part-of-speech (POS) ratios and tense features.
- **Syntax:** dependency tree depth.

5. Model

A **Random Forest Regressor** is used to predict the grammar score. Grammar scoring is treated as a **regression** problem rather than classification because scores take fractional values such as 2.5, 3.5 and 4.5, so the labels are ordered and continuous in nature. Random Forests handle mixed feature types well, capture non-linear relationships and are robust on small datasets.

6. Evaluation

The model was evaluated using RMSE (against a baseline and with 5-fold cross-validation) and Pearson correlation.

Metric	Result
Baseline RMSE	1.2382
5-Fold CV RMSE	0.7876
Pearson Correlation	[Add final value]

The following plots were also produced to analyse the model:

- **Actual vs Predicted plot** – compares predicted grammar scores with true scores.
- **Feature importance plot** – shows which features contribute most to the Random Forest.
- **Label distribution plot** – shows how grammar scores are distributed in the dataset.

7. Conclusion

Combining speech (audio), linguistic (text) and grammatical (POS, tense, dependency) features improved performance over the baseline, reducing RMSE from 1.2382 to 0.7876 under 5-fold cross-validation. This shows that grammar proficiency can be estimated effectively from spoken audio using a Whisper + Random Forest pipeline.